

1 Solution — GIAM 5.3

Problem 13

1.1 Problem

Give an inductive proof that $\forall x \geq 4 \in \mathbb{N}, x^2 2^x \leq 4^x$.

1.2 Answer

$$(k+1)^2 2^{k+1} \leq 2k^2 \cdot 2^{k+1} = 4 \cdot k^2 2^k \leq 4 \cdot 4^k = 4^{k+1}$$

The first inequality is the lemma $(k+1)^2 \leq 2k^2$, valid for $k \geq 3$; the second is the inductive hypothesis times 4. The basis $x = 4$ is an equality, $16 \cdot 16 = 256 = 4^4$, and the inequality is strict for every $x \geq 5$. Verified for every x from 4 to 3000.

1.3 The Proof

Lemma. For every natural number $k \geq 3$, $(k+1)^2 < 2k^2$.

Proof. $2k^2 - (k+1)^2 = k^2 - 2k - 1 = k(k-2) - 1$. For $k \geq 3$, $k(k-2) \geq 3$, so this is at least $2 > 0$. ■

Theorem. For every natural number $x \geq 4$, $x^2 2^x \leq 4^x$.

Proof. By induction on x , starting at 4.

Basis. At $x = 4$, $4^2 \cdot 2^4 = 16 \cdot 16 = 256$ and $4^4 = 256$, so $256 \leq 256$.

Inductive step. Let $k \geq 4$ be a particular but arbitrarily chosen natural number and suppose $k^2 2^k \leq 4^k$. By the lemma,

$$(k+1)^2 2^{k+1} < 2k^2 \cdot 2^{k+1} = 4 \cdot k^2 2^k.$$

By the hypothesis, $4 \cdot k^2 2^k \leq 4 \cdot 4^k = 4^{k+1}$. Chaining the two, $(k+1)^2 2^{k+1} < 4^{k+1}$, which in particular gives $\leq$.

Basis and step together give the result for every $x \geq 4$. ■

The step actually delivers strict inequality, so the theorem holds with $<$ from $x = 5$ on, and only the basis is an equality. That is why the problem uses $\leq$: with $<$, the claim would fail at $x = 4$.

Nor can the basis be moved earlier. The claim is false at $x = 3$ ($72 > 64$), and the lemma fails at $k = 2$ ($9 > 8$), so the step cannot cross from 2 to 3 either. The true cases $x = 0, 1, 2$ are cut off from the rest, and starting at 4, an equality, is the only way to reach the infinite tail.

1.4 The Picture

Dividing by 2^x reduces the claim to $x^2 \leq 2^x$, whose two curves are drawn on the right.

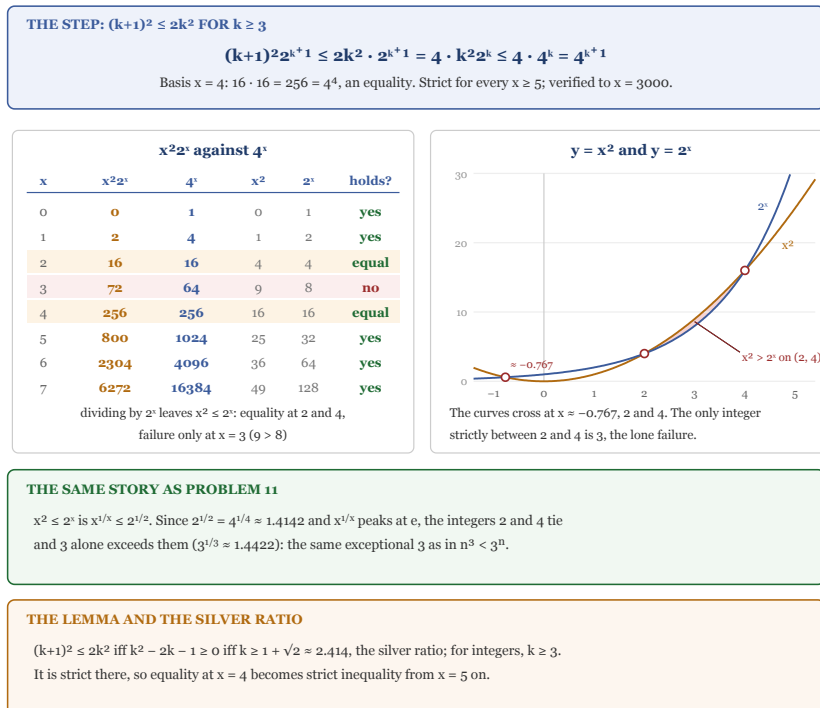


Figure 1.1: A diagram in four parts. The top band gives the step chain, using the lemma k plus one squared at most two k squared for k at least three. The left panel tabulates both sides for x from zero to seven: the inequality holds everywhere except x equal to three, with equality at two and four. The right

panel graphs x squared and two to the x , crossing at about minus 0.767, two and four, with x squared larger on the shaded interval from two to four. Two bands link the exception at three to Problem 11 and give the lemma's threshold, the silver ratio.

1.5 Remark — Really a Comparison of Squares and Powers of Two

Since $4^x = 2^x \cdot 2^x$, dividing by $2^x > 0$ gives

$$x^2 2^x \leq 4^x \iff x^2 \leq 2^x.$$

So the problem is the classic comparison of x^2 and 2^x in disguise, and the lemma $(k+1)^2 < 2k^2$ is exactly the ratio condition for it: one step multiplies 2^x by 2 but x^2 only by $(1+1/x)^2$. The lemma holds precisely when $k^2 - 2k - 1 \geq 0$, that is, $k \geq 1 + \sqrt{2} \approx 2.414$. The number $1 + \sqrt{2}$ is the *silver ratio*, the companion of the golden ratio of Problem 12.

Over the reals, $x^2 = 2^x$ at $x \approx -0.767$, $x = 2$ and $x = 4$, and $x^2 > 2^x$ exactly on $(2, 4)$ among positive x . Since 3 is the only integer strictly between 2 and 4, it is the only natural number where the inequality fails: $9 > 8$. Every natural number other than 3 satisfies it, checked for x from 0 to 199.

1.6 Remark — The Same Exceptional 3 as Problem 11

Taking x -th roots, $x^2 \leq 2^x$ becomes $x^{1/x} \leq 2^{1/2}$. In Problem 11 we saw that $x^{1/x}$ rises to a peak at $x = e$ and then falls. Near the peak, the integers 2, 3 and 4 give

$$2^{1/2} = 4^{1/4} \approx 1.4142, \quad 3^{1/3} \approx 1.4422.$$

The equality $2^{1/2} = 4^{1/4}$ is just $2^4 = 4^2 = 16$, the famous coincidence behind the two equality cases. Only $3^{1/3}$ is larger, so $x = 3$ is the lone failure, exactly as it was the lone failure of $n^3 < 3^n$. Both problems are reading off the same fact: among natural numbers, $n^{1/n}$ is largest at $n = 3$.